AI Resume Analyzer and Job Recommendation System

Project Overview

This project is a Python-based application that analyzes PDF and DOCX resumes, identifies technical skills, and recommends relevant job roles using keyword matching and text similarity.

Problem Statement and Objectives

Job seekers may find it difficult to identify suitable career paths and the skills required for target roles. The system aims to simplify this process by extracting resume information, comparing skills with predefined job requirements, and highlighting potential learning gaps.

Methodology

1. Upload a PDF or DOCX resume.
2. Extract and clean the resume text.
3. Detect technical skills using regular expressions and a skill dictionary.
4. Compare the resume with six predefined job roles using TF-IDF and cosine similarity.
5. Calculate skill coverage, rank recommendations, and display missing skills for the selected role.

Key Features

Resume upload and parsing; technical skill detection; top-three job recommendations; skill coverage charts; matched and missing skill analysis; and an interactive Streamlit interface.

Technology Stack

Python, Streamlit, Pandas, NumPy, pypdf, python-docx, Scikit-learn, TF-IDF, cosine similarity, and Matplotlib.

Testing and Limitations

The application has been tested with a sample DOCX resume, and the main analysis results were displayed successfully. Additional PDF and edge-case tests should be recorded separately. Recommendations depend on predefined roles and keywords; scanned PDFs may require OCR. Scores are estimates and should not be treated as hiring decisions.

Conclusion

The project demonstrates how NLP-related text processing and machine learning similarity techniques can support career exploration and personalized skill development.